

1. Neural Networks (Fundamentals)

1.1 Introduction

- Biological inspiration of neural networks
- Artificial neuron
- Neural network architectures

1.2 Mathematical Model of a Neuron

- Weighted sum and bias
- Activation functions:
 - Sigmoid
 - Tanh
 - ReLU
 - Softmax

1.3 Network Architectures

- Single-layer perceptron
- Multi-layer perceptron (MLP)
- Feedforward vs feedback networks

1.4 Learning Process

- Loss functions (MSE, Cross-Entropy)
- Backpropagation algorithm
- Learning rate

1.5 Practical Issues

- Vanishing and exploding gradients
 - Overfitting and underfitting
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2. Convolutional Neural Networks (CNN)

2.1 Motivation for CNN

- Limitations of fully connected networks for images
- Local connectivity and weight sharing

2.2 CNN Architecture

- Convolution layer
 - Kernel/filter
 - Stride and padding
 - Channels
- Activation functions after convolution
- Pooling layer (Max, Average)
- Fully connected layer

2.3 Feature Learning in CNN

- Edge detection in early layers
- Texture and shape learning
- End-to-end learning

2.4 Training CNN

- Batch size and epochs
- Batch Normalization
- Data augmentation

2.5 Applications

- Image classification
- Object detection
- Face recognition
- Medical imaging
- Sentiment Classification

3. Natural Language Processing (NLP)

3.1 Introduction to NLP

- What is NLP?
- Challenges in NLP
- Applications (translation, chatbots, sentiment analysis)

3.2 Text Preprocessing

- Tokenization
- Stop-word removal
- Stemming and lemmatization

3.3 Text Representation

- TF-IDF
- Word embeddings (Word2Vec)

3.4 Neural Models for NLP

- RNN for sequence modeling

3.5 NLP Tasks

- Text classification
 - Language modeling
 - Machine translation
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4. Recommendation Systems

4.1 Introduction

- What is a recommendation system?
- Importance in real-world applications

4.2 Types of Recommendation Systems

- Content-based filtering
- Collaborative filtering
 - User–user
 - Item–item
- Hybrid recommendation systems

4.3 Collaborative Filtering Techniques

- Similarity measures:
 - Cosine similarity
 - Pearson correlation
- Matrix factorization
- Cold start problem

4.5 Evaluation Metrics

- Precision
- Recall
- RMSE
- MAE

5. Ethical and Practical Issues

- Bias in neural and recommendation systems
- Data imbalance
- Model interpretability
- Privacy concerns